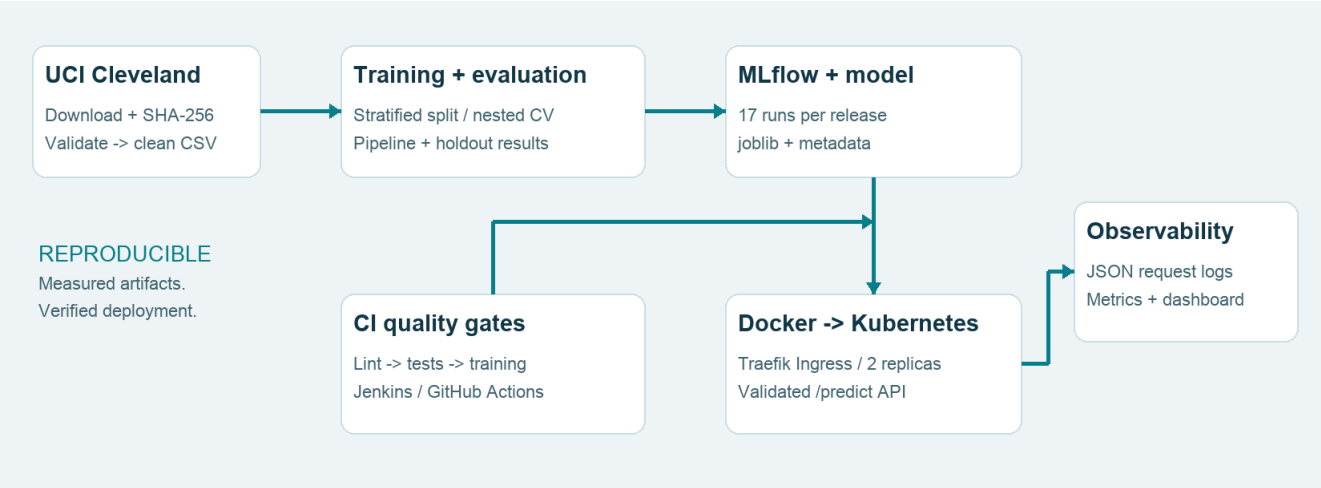


01 Heart disease prediction

An end-to-end MLOps implementation using the UCI Cleveland dataset: reproducible data preparation, leakage-aware experiments, a packaged prediction API, automated CI, container execution and a monitored local Kubernetes deployment.

Ankit Kumar Agrawal | 2025AE05620

Release outcome	Measured evidence
Selected model	Logistic regression; nested CV AUC 0.9151 +/- 0.0318
Untouched holdout	61 rows; accuracy 0.8033; recall 0.8214; ROC-AUC 0.9232
Validation	26 pytest cases passed; independent release verification passed
Serving	Non-root Docker API; 2 ready Kubernetes replicas; Traefik ingress



Scope and submission location

This is an educational classifier for the recorded presence of heart disease. It is not a prospectively validated clinical risk tool. The delivered artifact is the exact training-partition model evaluated on holdout; the holdout is not reused for refitting.

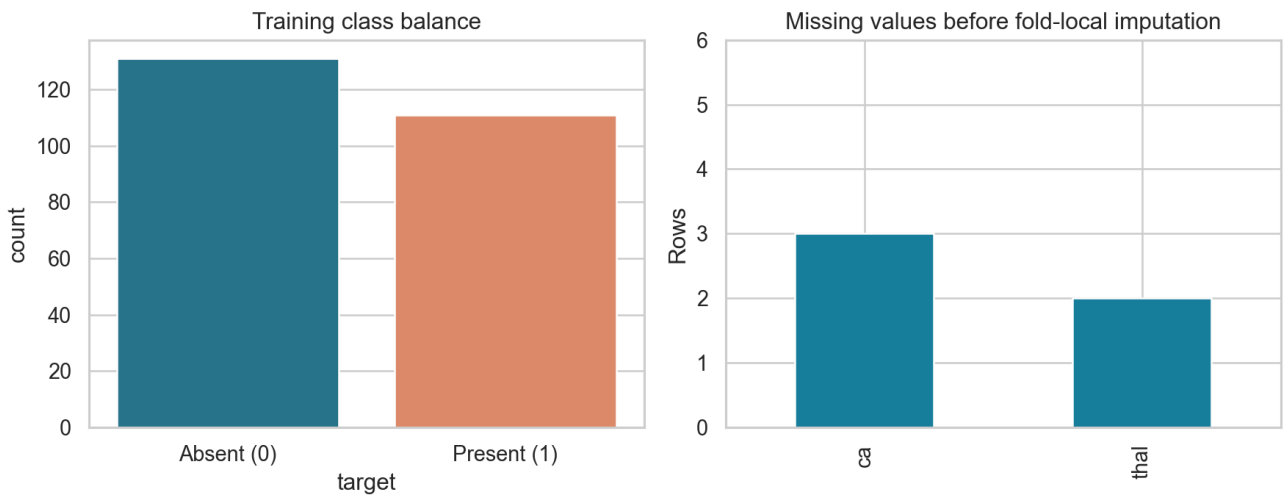
Repository publication pending: GitHub account authentication and the repository destination were not supplied. All source is in the accompanying mlops-heart project. Add the real repository URL before submission; a local path is not a GitHub link.

Local deployed access: <http://127.0.0.1:8080/docs>. Reproduction instructions and the video are included. Screenshots document actual execution, with rendered transcripts explicitly labelled.

02 Data acquisition & quality

Source: Janosi, Steinbrunn, Pfisterer and Detrano (1989), UCI Heart Disease, DOI 10.24432/C52P4X. The official processed Cleveland file has 303 records and 13 predictors plus the target. This explains the assignment's shorthand "14+ features": the canonical table has 14 columns, not 14 independent predictors. Original UCI category codes are preserved.

Group	Variables and original coding
Continuous	age (years), trestbps (mm Hg), chol (mg/dL), thalach (bpm), oldpeak (ST depression)
Categorical	sex 0/1; cp 1-4; fbs 0/1 (>120 mg/dL); restecg 0-2; exang 0/1; slope 1-3; ca 0-3; thal 3/6/7
Target	num=0 -> absent (0); num=1,2,3,4 -> present (1)
Quality checks	303 retained rows; no exact duplicates; 4 ca and 2 thal values missing; 164 negative / 139 positive



Repeatable cleaning, without leakage

The acquisition script fetches the official archive or uses the included raw file, verifies its SHA-256, validates columns, target values and category domains, normalizes missing markers and removes exact duplicate rows. The cleaned CSV retains nulls because imputation is learned separately within each CV training fold. The manifest records data origin, hash, schema and quality counts.

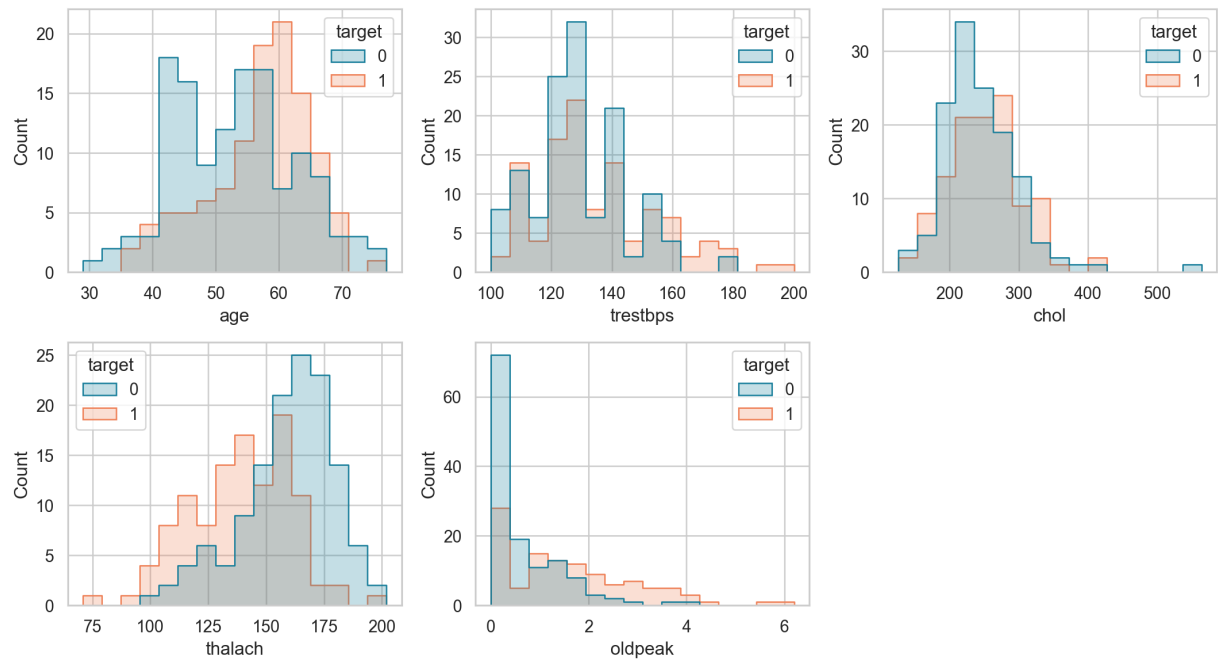
```
python -m heart.data
# outputs: data/processed/heart.csv and manifest.json
```

Raw SHA-256: a74b7efa387bc9d108d7d0115d831fe9b414b29ae7124f331b622b4efa0427c8. The downloader fails if the raw content changes, rather than silently changing the experiment. UCI identifies the dataset license as CC BY 4.0.

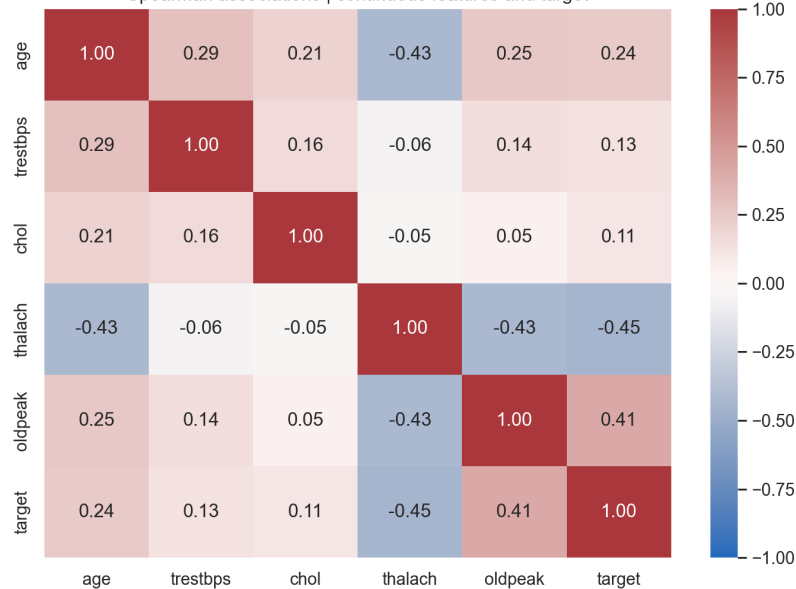
03 Exploratory analysis

A stratified 80/20 split (seed 523) is made before exploratory plotting. The training partition contains 242 records: 131 negative and 111 positive. The test partition remains separate. Mild class imbalance supports stratified folds and explicit recall/precision reporting rather than synthetic resampling.

Clinical distributions | training partition only (n=242)



Spearman associations | continuous features and target



Histograms show continuous distributions by class, including overlapping groups and right-skewed oldpeak. The Spearman heatmap is restricted to continuous features and the binary target: numeric codes for nominal categories are not interpreted as distances. Observational associations are not causal effects. Extreme values are retained because this small dataset offers no justified automatic clinical outlier-removal rule.

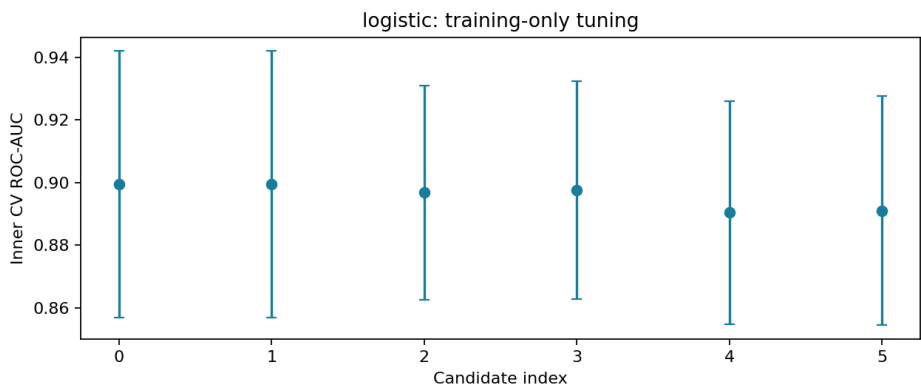
04 Features, tuning & selection

One sklearn Pipeline includes a ColumnTransformer and classifier. Numeric inputs receive median imputation and StandardScaler; categorical inputs receive most-frequent imputation and OneHotEncoder with explicit original UCI categories. Encoding is nominal, including ca and slope, to avoid assuming equal spacing. This specification is shared by training and inference.

Model	Prespecified search space
Logistic regression	C = 0.1, 1, 10; class_weight = None or balanced; max_iter=2000
Random forest	200 trees; max_depth = 3 or None; min_samples_leaf = 2 or 5; class_weight = None or balanced

Each family is evaluated using five stratified outer folds; a four-fold inner GridSearchCV tunes ROC-AUC. All learned preprocessing is refitted inside the inner folds. The final family is selected by mean outer AUC, then its grid search is refitted on all 242 training rows. Threshold 0.5 is specified in advance. The holdout never selects hyperparameters or threshold.

Outer-CV mean +/- sample SD	Logistic regression	Random forest
ACCURACY	0.8513 +/- 0.0448	0.8390 +/- 0.0329
PRECISION	0.8616 +/- 0.0921	0.8393 +/- 0.0852
RECALL	0.8186 +/- 0.1023	0.8190 +/- 0.0731
ROC AUC	0.9151 +/- 0.0318	0.9118 +/- 0.0283

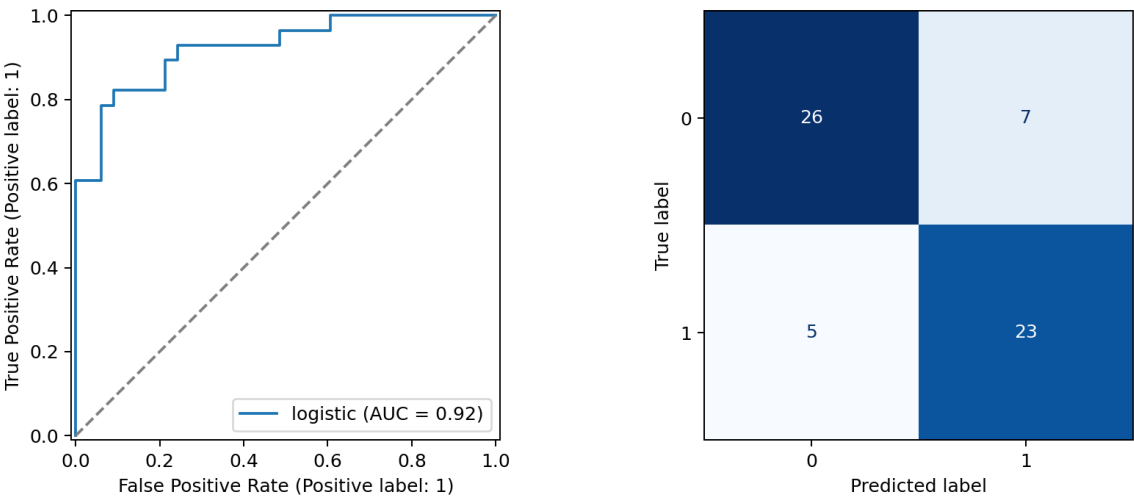


Logistic regression wins by a small mean-AUC margin (0.0033); fold variability is much larger, so this is a deterministic selection rule, not proof of statistical superiority. The simpler model is also easier to inspect. Selected logistic parameters: {'classifier__C': 0.1, 'classifier__class_weight': None}.

05 Final evaluation & limitations

The selected model was evaluated on the 61 held-out records only after model-family selection. The persisted split and row-level predictions support independent recomputation. The saved serving model is the same fitted object used here.

Metric	Holdout value	Interpretation
Accuracy	0.8033	49 of 61 correctly classified
Precision	0.7667	23 of 30 predicted-positive cases were positive
Recall	0.8214	23 of 28 positive cases detected
F1	0.7931	Balance of precision and recall
ROC-AUC	0.9232	Ranking quality across thresholds



Uncertainty and operational meaning

A 2,000-resample bootstrap on the held-out predictions gives a descriptive 95% AUC interval of [0.847, 0.977]. This reflects resampling of this small holdout, not all model-selection uncertainty. Confusion counts are TN=26, FP=7, FN=5, TP=23. False negatives matter in screening, but changing the threshold requires training-only validation and a documented cost trade-off.

The small historical cohort may not represent current or diverse patient populations. There is no external validation, calibration study, subgroup guarantee or longitudinal outcome validation. Confidence is the model's returned-class probability, not a guaranteed probability of disease in a new population. Clinical use would require independent validation and governance.

06 Experiment tracking

MLflow captures a release parent, two family runs and fourteen candidate runs (six logistic, eight forest): 17 completed runs per release. Candidate records log parameters, fold AUC values, metric means and standard deviations, a JSON artifact and tuning plot. Family runs retain complete CV tables, selected hyperparameters and nested-CV summaries.

The screenshot shows the MLflow Experiments page for an experiment named 'reproducible-comparison'. The interface includes a navigation bar with 'Experiments', 'Models', and 'Prompts'. Below the navigation bar, there are tabs for 'Overview', 'Model metrics', 'System metrics', 'Traces', and 'Artifacts'. The 'Overview' tab is selected, displaying a table of experiment details and two sections for parameters and metrics.

Experiment ID	881943188350597665
Status	Finished
Run ID	687db1f906644749b541ee8a43a47528
Duration	27.8s
Datasets used	—
Tags	Add tags
Source	train.py
Logged models	sklearn
Registered models	—
Registered prompts	—

Parameters (8)

Parameter	Value
test_rows	61
seed	523
inner_folds	4
winner	logistic
outer_folds	5
threshold	0.5
train_rows	242
data_sha256	e81973d3356c231ed8903aab16f5faa72d9679eb8ef6dc98a602b25df3f5d23d

Metrics (5)

Metric	Value
holdout_recall	0.8214285714285714
holdout_accuracy	0.8032786885245902
holdout_precision	0.7666666666666667
holdout_roc_auc	0.9231601731601732
holdout_f1	0.7931034482758621

What can be inspected

The parent logs dataset hash, random seed, partition sizes, fold counts and threshold; holdout metrics; EDA and evaluation plots; exact split and predictions; and an MLflow sklearn model with signature and example. The experiment export in reports/experiments.csv and .json contains the final release run tree. Earlier development runs are distinguishable by run ID.

```
python scripts/relocate-mlflow.py # only after moving this directory
mlflow ui --backend-store-uri ./mlruns --host 127.0.0.1 --port 5001
# Open :5001, then select heart-disease-cleveland
```

Selected release run ID: 687db1f906644749b541ee8a43a47528. Artifact paths in MLflow's local file store are absolute, so relocation explicitly updates them. The portable JSON/CSV exports remain readable without a tracking server.

07 Packaging & API contract

The reusable artifact includes all fitted transforms plus the classifier in models/heart_pipeline.joblib. Metadata stores features, threshold, model/data hashes, split sizes and run ID. API startup verifies SHA-256 before loading. It fails closed on missing or mismatched artifacts; no serving-time training or data download occurs.

```
python3 -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt -e .
python -m heart.data && python -m heart.eda && python -m heart.train
pytest -q
python scripts/verify-release.py
bash scripts/container-smoke.sh
```

requirements.txt pins training/testing dependencies; requirements-serve.txt is a smaller serving set. The Docker image uses Python 3.13.3, runs as non-root UID 10001 and includes the model. The isolated smoke test uses a read-only filesystem, no Linux capabilities and no host code/model mount. Build and real response proof are saved in evidence/docker.log.

Prediction request and measured response

```
POST /predict
{"age":63,
 "sex":1,
 "cp":1,
 "trestbps":145,
 "chol":233,
 "fbs":1,
 "restecg":2,
 "thalach":150,
 "exang":0,
 "oldpeak":2.3,
 "slope":3,
 "ca":0,
 "thal":6}
```

Response field	Observed value
prediction / threshold	0 / 0.5
disease_probability / confidence	0.432392 / 0.567608
model_version	074c8b3372d9

Pydantic rejects unexpected fields, invalid categories, out-of-range values and missing required fields with HTTP 422. Declared nullable fields use the fitted imputers. For a positive prediction confidence=p; for a negative prediction confidence=1-p. /health is liveness, /ready confirms loaded model, /docs supplies interactive OpenAPI documentation.

08 Automated CI & testing

The repository contains both GitHub Actions YAML and the allowed Jenkins alternative. The actual local Jenkins run uses a clean Linux virtual environment to install pinned dependencies, lint, validate data, run unit tests, train tracked models, test the API and verify model provenance. Tests and artifacts are retained. Jenkins has no host Docker socket access.

 Jenkins

heart-mlops

#3

Tests

Status

Changes

Console Output

Delete build '#3'

Tests

Edit build information

Restart from Stage

Replay

Pipeline Steps

Workspaces

Previous Build

Tests 26

✓ 26 Took 1.1 sec

All tests are passing!

Nice one! All 26 tests are passing.

All Tests

Package	Failed	Skipped	Passed	Total	Duration
test	0	0	26 +26	26 +26	0.45 sec

REST API Jenkins 2.580.1

Test coverage	Assertions
Data processing (13)	Target mapping, missing values, deduplication, invalid categories/targets/ranges, schema drift
Model pipeline (3)	Both families impute, probability sums, serialization round-trip and fit-only imputation
API (10)	Prediction semantics, metrics/health, extra/invalid/missing fields, nullable values, checksum rejection

Local pytest result: 26 passed. The workflow fails on lint or test failures; bash pipefail preserves failures through logging pipes. GitHub Actions additionally builds/tests the container and creates an ephemeral kind cluster to verify ingress deployment. Its artifacts are uploaded even on failure. A GitHub Actions run is not claimed without repository authentication; the screenshot above is Jenkins evidence.

Docker and Kubernetes were also verified directly on this machine, independently of Jenkins. CI retraining may serialize a different model checksum across Python/OS versions; each job verifies its own newly generated artifact and reports the resulting metrics.

09 Kubernetes deployment

The actual deployment uses a dedicated kind cluster on Colima. `k8s/app.yaml` defines the namespace, two-replica Deployment, Service and Ingress; `k8s/ingress-controller.yaml` defines Traefik. Host port 8080 maps to NodePort 30080, then the Ingress routes through the Service to ready API pods. No public cloud URL is claimed.

AIMLCZG523 / MLOPS ASSIGNMENT 01 / MEASURED EVIDENCE

Kubernetes deployment verification

[Overview](#) [EDA](#) [Models](#) [Deployment evidence](#)

Rendered transcript of actual saved command output. This is an evidence viewer, not a Kubernetes UI.

kubectl --context kind-heart-mlops -n heart-mlops get deploy,pods,svc,ingress -o wide

NAME	READY	UP-TO-DATE	AVAILABLE	AGE	CONTAINERS	IMAGES	SELECTOR
deployment.apps/heart-api	2/2	2	2	76s	api	heart-api:local	app=heart-api
deployment.apps/traefik	1/1	1	1	75s	traefik	traefik:v3.3.6	app=traefik

NAME	READY	STATUS	RESTARTS	AGE	IP	NODE	NOMINATED NODE
READINESS GATES							
pod/heart-api-85488ddf6d-g5lvr	1/1	Running	0	76s	10.244.0.6	heart-mlops-control-plane	<none>
pod/heart-api-85488ddf6d-kt868	1/1	Running	0	76s	10.244.0.5	heart-mlops-control-plane	<none>
pod/traefik-8684b74f5f-vn2kr	1/1	Running	0	75s	10.244.0.7	heart-mlops-control-plane	<none>

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE	SELECTOR
service/heart-api	ClusterIP	10.96.119.10	<none>	80/TCP	76s	app=heart-api
service/traefik	NodePort	10.96.92.84	<none>	80:30080/TCP	75s	app=traefik

NAME	CLASS	HOSTS	ADDRESS	PORTS	AGE
ingress.networking.k8s.io/heart-api	traefik	*		80	76s

POST http://127.0.0.1:8080/predict - actual API response

```
{
  "prediction": 0,
  "disease_probability": 0.432391735603244,
  "confidence": 0.5676082643967559,
  "model_version": "074c8b3372d9",
  "threshold": 0.5
}
```

Request path: localhost:8080 → Traefik ingress → Service → 2 API replicas.

```
docker build -t heart-api:local .
bash scripts/deploy-local.sh
curl -fsS http://127.0.0.1:8080/ready
python scripts/infer.py --url http://127.0.0.1:8080
kubectl --context kind-heart-mlops -n heart-mlops get pods,ingress
```

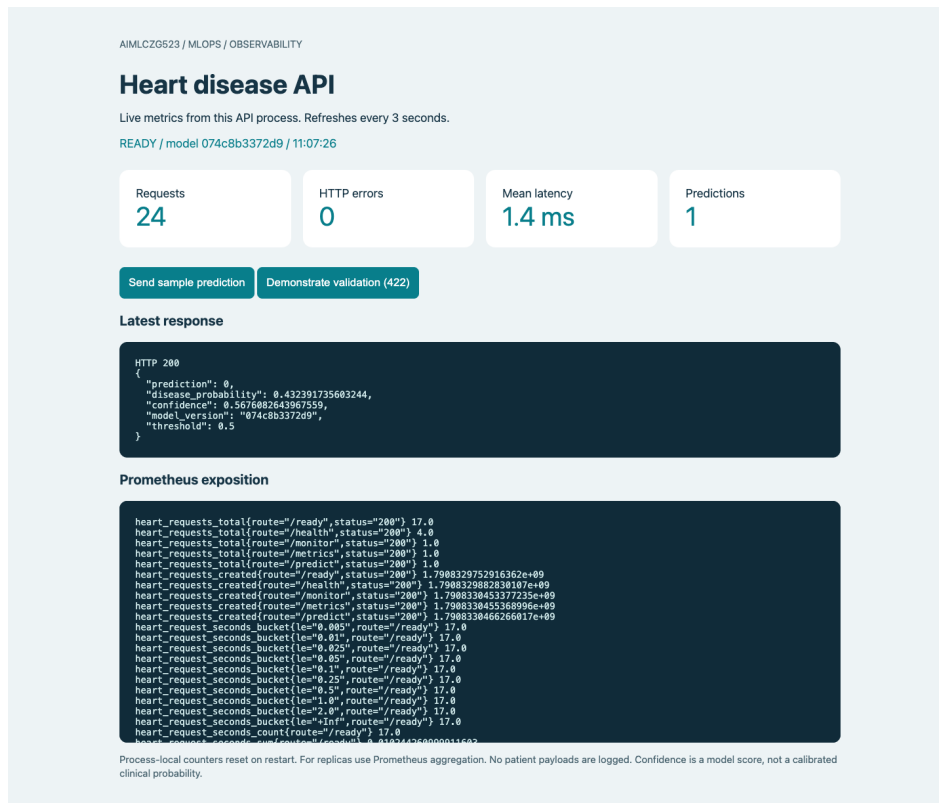
Availability and release control

Startup/readiness/liveness probes prevent premature traffic and detect failures. Each API pod requests 100m CPU / 256Mi RAM and is limited to 1 CPU / 768Mi RAM. Pods run non-root with read-only root filesystems and dropped capabilities. Rolling updates set `maxUnavailable=0` and `maxSurge=1`. Logs confirm real requests and the reported model version.

For repeat releases use unique image tags, load the image into kind and update the Deployment. Reusing a local tag requires an explicit rollout restart. Rollback: `kubectl --context kind-heart-mlops -n heart-mlops rollout undo deployment/heart-api`. Verify readiness and sample inference after changes. These controls demonstrate cloud-ready packaging; TLS, authentication, persistent monitoring and a managed registry remain production extensions.

10 Monitoring & handoff

Request middleware emits JSON request ID, method, bounded route label, status and duration without patient payloads. Prometheus counters and histograms back the live /monitor dashboard. Successful inference and HTTP 422 rejection are demonstrated; screenshots and raw logs are included.



The dashboard displays one process at a time; with two replicas its counters are not a cluster-wide total and reset on restart. Production Prometheus should scrape every pod and aggregate. monitoring/README.md supplies example throughput, error-rate and p95 expressions. Accuracy/drift monitoring requires ground-truth labels and is not claimed by request metrics alone.

Deliverables and final submission check

Included: source, cleaned/raw data and downloader, EDA/training/inference scripts, pinned dependencies, serialized model, 26 tests, Dockerfile, GitHub Actions YAML, Jenkinsfile, manifests, execution logs, screenshots, MLflow runs/exports, this 10-page report and reports/pipeline-walkthrough.mp4. README.md gives complete setup, API access, restart and cleanup instructions. Publish the GitHub repository and add its real link and student details before submission if they are still absent.

Sources and implementation references

UCI Heart Disease: doi.org/10.24432/C52P4X. scikit-learn nested CV: [official nested cross-validation example](#). MLflow: [tracking documentation](#). kind: [cluster and port-mapping configuration](#). Assignment specification: supplied MLOps Assignment 1 Instructions.pdf. All numeric results and operational evidence in this report come from the accompanying executions.